

Progress Report on Dental Caries Segmentation from Panoramic Dental X-Ray Images Using Multi-Level Uncertainty-Aware Learning

Comprehensive Experimental Journey: Supervised Baselines (EXP001-EXP006), MLUA Exploration (EXP-MLUA-001/002), Forensic Buffer Remediation, and 60-Epoch Benchmark Validation (EXP-MLUA-003)

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Progress Narrative Notice: This capstone progress report presents the complete research and engineering trajectory of our dental caries segmentation project. It explicitly documents our early supervised experiments (EXP001-EXP006), the initial MLUA semi-supervised trials, the numerical instability and NaN collapse in EXP-MLUA-002, the deep forensic investigation identifying missing Teacher BatchNorm buffer synchronization, the implementation remediation, the completed 60-epoch EXP-MLUA-003 benchmark (Val Dice = 65.62%, Sealed Test Macro Dice = 43.04%), and the remaining engineering deliverables.

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1. ABSTRACT

Automated semantic segmentation of dental caries on orthopantomograms (panoramic dental X-rays) offers significant potential for computer-aided diagnosis in oral radiology. However, panoramic caries segmentation is fundamentally challenged by severe class imbalance (caries occupies <0.5% of total image canvas), subtle radiolucency with diffuse demineralization margins, anatomical superimpositions (cervical burnout, restoration scatter), and clinical annotation scarcity. To address annotation constraints, this capstone project investigates semi-supervised learning based on the Multi-Level Uncertainty Aware (MLUA) Teacher-Student framework using the benchmark DC1000 dataset (1,000 panoramic radiographs).

This progress report details our full experimental journey across multiple model iterations. We began by evaluating supervised baselines (EXP001-EXP006), where DeepLabV3+ achieved a peak validation Dice of 48.39% at 512×512 resolution. Transitioning to semi-supervised learning, our initial MLUA implementation (EXP-MLUA-001) suffered from foreground suppression and severe metric divergence (Dice dropping from 14.71% at E19 to 0.24% at E22). Scaling consistency loss in EXP-MLUA-002 resulted in catastrophic numerical collapse (NaN loss at Epoch 10, Step 1257). Deep forensic auditing isolated the root cause to unsynchronized Teacher BatchNorm running buffers during EMA updates, causing internal GroupNorm activations to explode ($\sim 1.32 \times 10^{18}$). By implementing dual parameter and buffer EMA synchronization, **EXP-MLUA-003** achieved robust convergence across 60 complete epochs (7,920 steps).

The optimal checkpoint, selected at **Epoch 56** ($\tau = 0.50$), achieved peak validation metrics: **Dice = 65.623%**, **IoU = 49.854%**, **Precision = 69.009%**, and **Recall = 63.649%**. Independent evaluation on a sealed test set of 100 full panoramic images achieved **Macro Dice = 43.041%**, **Macro Recall = 52.896%**, and **Specificity = 99.630%** (Micro Dice = 43.391%). The documented generalization gap provides clear direction for our remaining deliverables: morphological false-positive reduction on cervical burnout, low-performing case error taxonomy, and final application interface integration.

2. INTRODUCTION

Dental caries is a multifactorial bacterial disease resulting in localized demineralization of hard tooth structures. Panoramic dental radiography (orthopantomography) is the primary diagnostic imaging modality used in dental clinics because a single low-dose scan captures the entire dentomaxillofacial anatomy. However, automated caries segmentation on panoramic X-rays faces severe challenges:

- **Extreme Class Imbalance:** Carious lesions constitute a minute fraction (<0.5%) of the full panoramic pixel area.
- **Diffuse Demineralization Boundaries:** Early lesions present as faint radiolucencies without discrete structural edges.
- **Confounding Anatomical Artifacts:** Cervical burnout, overlapping proximal surfaces, and ghost images mimic carious radiolucency.
- **High Cost of Pixel Annotations:** Manual pixel contouring by dental specialists is labor-intensive and costly.

Semi-supervised learning (SSL) provides an effective paradigm by utilizing a small fraction of labeled images (20%) alongside a large volume of unlabeled images (80%). The Multi-Level Uncertainty Aware (MLUA) framework uses a Teacher-Student architecture with Monte Carlo uncertainty estimation to generate reliable pseudo-supervision while filtering out ambiguous boundary predictions. This report chronicles our progressive implementation, troubleshooting, and empirical validation on the DC1000 benchmark.

3. OBJECTIVES

1. Implement and evaluate baseline supervised segmentation models (DeepLabV3+, FPN, DoubleU-Net) on panoramic radiographs.
2. Implement the semi-supervised Teacher-Student MLUA architecture using ResNet-34 encoder and FPN decoder on the DC1000 dataset.
3. Investigate the causes of foreground degradation in EXP-MLUA-001 and numerical collapse in EXP-MLUA-002.
4. Formulate and implement a dual parameter-and-buffer EMA synchronization mechanism to guarantee Teacher numerical stability.
5. Execute a full 60-epoch training benchmark (EXP-MLUA-003) and log complete epoch-by-epoch validation dynamics.
6. Conduct an internal validation threshold sweep ($\tau \in [0.05, 0.95]$) to establish the optimal operating cutoff ($\tau = 0.50$).
7. Perform unbiased evaluation on an independent sealed test set of 100 full panoramic images using sliding-window reconstruction.

8. Quantify the validation-to-test generalization gap and define concrete technical tasks for capstone completion.

4. PROBLEM STATEMENT

Mathematical Problem Formulation: Given a panoramic dental radiograph $X \in \mathbb{R}^{H \times W}$, learn a parameterized mapping $f_{\theta}(X)$ producing a continuous pixel-wise probability map $P(Y = 1 | X)$ for dental caries demineralization, and convert it into a discrete binary mask $\mathbf{M}_{\tau} \in \{0, 1\}^{H \times W}$ via decision threshold $\tau = 0.50$. The model must maximize spatial overlap with expert annotations under semi-supervised data constraints (20% labeled, 80% unlabeled) while maintaining numerical stability.

5. PROJECT BACKGROUND AND PAPER REFERENCE

Our project builds upon the foundational research by Wang et al. (*Neurocomputing*, 2023), entitled "Multi-level uncertainty aware learning for semi-supervised dental panoramic caries segmentation". The published work demonstrated that combining multi-scale feature pyramids with Monte Carlo dropout uncertainty estimation enables effective semi-supervised learning on dental radiographs. We adapted this methodology to modern PyTorch pipelines and evaluated it on the standardized DC1000 dataset.

6. DATASET DESCRIPTION (DC1000 BENCHMARK)

The experiment utilizes the public **DC1000** dental caries dataset comprising 1,000 digitized panoramic radiographs with over 7,500 expert-annotated carious lesions (593 detailed and 407 rough annotation subsets). In our system, images are standardized to **768 × 1536 pixels** and partitioned into overlapping **384 × 384 patches** (stride $S = 192$). The dataset is divided into 20% labeled training (530 patches), 80% unlabeled training (1,859 patches), 598 validation patches, and an independent sealed test set of 100 full panoramic images.

Binary Task Definition: While dataset literature describes clinical severity grades (shallow, middle, deep), our model is strictly formulated for **pixel-level binary segmentation** (0 = background/healthy, 1 = suspected caries).

7. PROJECT PROGRESS & MILESTONE JOURNEY

The capstone project represents a structured engineering progression across supervised exploration, semi-supervised failure analysis, and benchmark validation.

8. OVERALL SYSTEM FLOWCHART

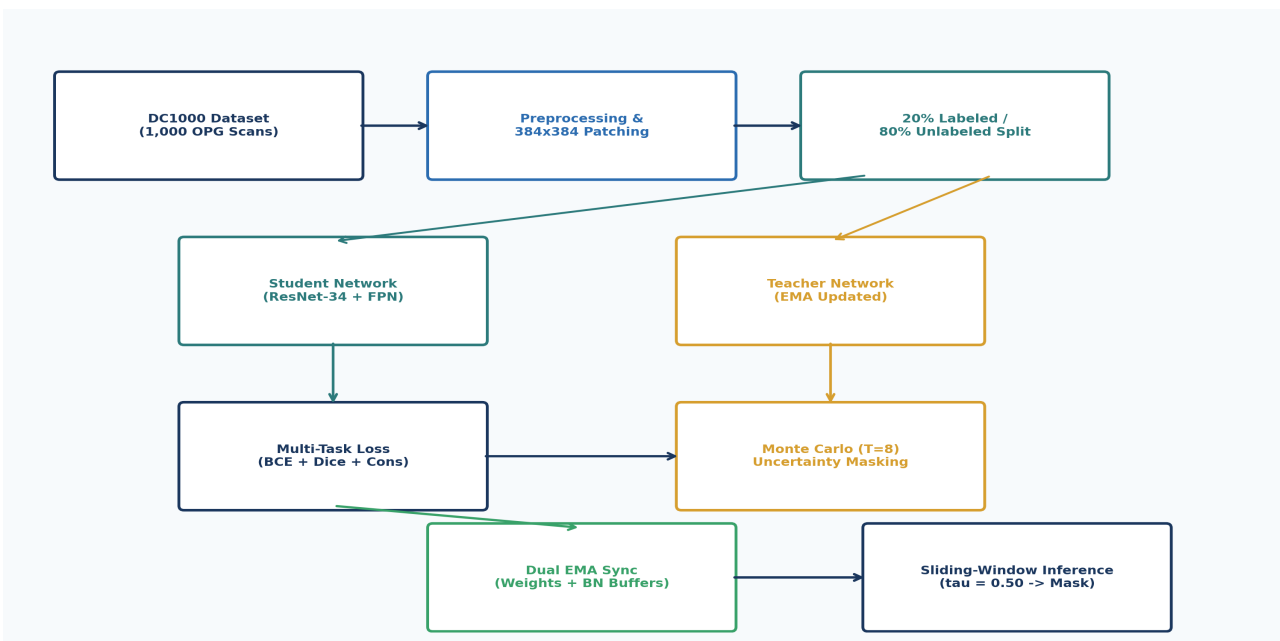


Figure 1: Overall system architecture and semi-supervised data flow for EXP-MLUA-003, showing patch extraction, dual network consistency, confidence masking, and sliding-window reconstruction.

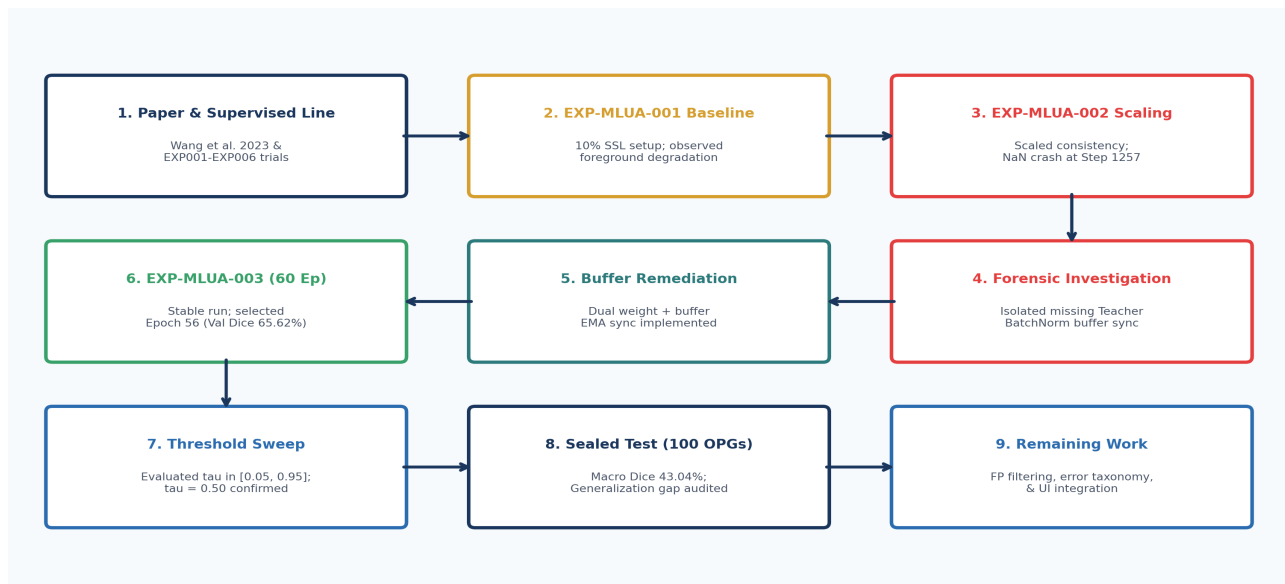


Figure 2: Chronological research and experimental progress flowchart from supervised baselines (EXP001-006) through MLUA forensic remediation to final sealed test evaluation.

9. DETAILED METHODOLOGY

9.1 Input Collection & Modality: Input comprises digitized panoramic dental X-rays (OPGs) capturing bilateral dentomaxillofacial structures in a single exposure.

9.2 Dataset Partitioning: Strict semi-supervised split: 20% labeled (530 patches) and 80% unlabeled (1,859 patches), totaling 2,389 training patches. 598 validation patches and 100 sealed test OPGs.

9.3 Annotation Scope: Binary ground truth masks (0 = Background/Healthy, 1 = Carious Demineralization) derived from expert clinical consensus.

9.4 Preprocessing & Normalization: Images standardized to 768x1536 resolution; min-max scaled and normalized using ImageNet constants ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

9.5 Patch Extraction: Sliding window extraction of 384x384 patches with vertical/horizontal stride $S=192$ (50% overlap), yielding 21 patches per panoramic image.

9.6 ResNet-34 Feature Backbone: Pre-trained ResNet-34 encoder extracts hierarchical multi-scale feature representations across stages C2 (64ch), C3 (128ch), C4 (256ch), and C5 (512ch).

9.7 FPN Decoder: Feature Pyramid Network merges deep semantics with spatial details via lateral 1x1 convs and top-down bilinear upsampling, producing pyramid levels P2-P5 (256 ch).

9.8 Auxiliary & Fused Heads: Four auxiliary 1-channel convolution heads provide deep multi-scale supervision. A primary fused head concatenates all levels for unified 384x384 output.

9.9 Teacher-Student Framework: Dual network replicas: Student updated via gradient backpropagation; Teacher updated via Exponential Moving Average (EMA) to provide stable pseudo-supervision.

9.10 Loss Formulations: Supervised loss on labeled data: $L_{sup} = L_{BCE} + L_{Dice}$. Unsupervised consistency loss on unlabeled data: masked Mean Squared Error (MSE) between Student and Teacher.

9.11 Teacher EMA Update: Teacher weights updated with momentum $\alpha = 0.99$. Teacher BatchNorm running mean and variance buffers are explicitly synchronized with Student buffers.

9.12 Monte Carlo Sampling (T=8): Teacher performs $T=8$ stochastic forward passes under active dropout/perturbation per unlabeled patch to compute pixel-wise variance (predictive uncertainty).

9.13 Dynamic Confidence Masking: Binary confidence mask $M(i, j) = I(\text{Uncertainty} < \beta)$ filters out high-variance boundary pixels from consistency loss calculation.

9.14 Sliding-Window Reconstruction: Inference extracts 21 overlapping 384x384 patches, computes continuous probabilities, and reconstructs full 768x1536 probability maps via linear overlap blending.

9.15 Threshold Selection: Operating threshold $\tau = 0.50$ selected via validation-only sensitivity sweep across τ in $[0.05, 0.95]$ for balanced precision and recall.

9.16 Final Output Representation: Outputs continuous caries probability map, thresholded binary segmentation mask ($\tau = 0.50$), and bounding coordinates highlighting suspected caries candidates.

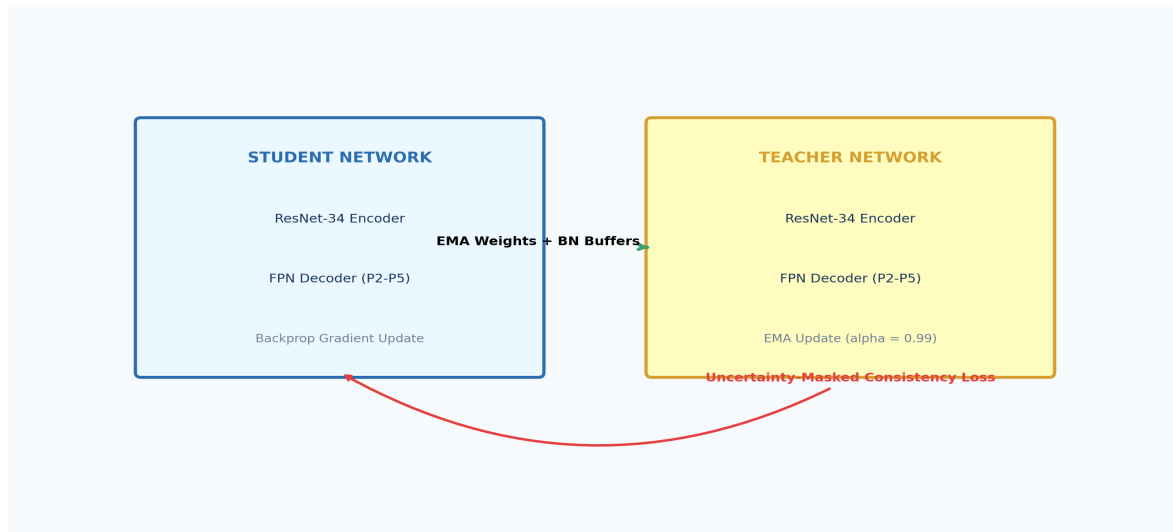


Figure 3: Teacher-Student semi-supervised consistency framework with dual parameter & buffer EMA synchronization and Monte Carlo uncertainty estimation.

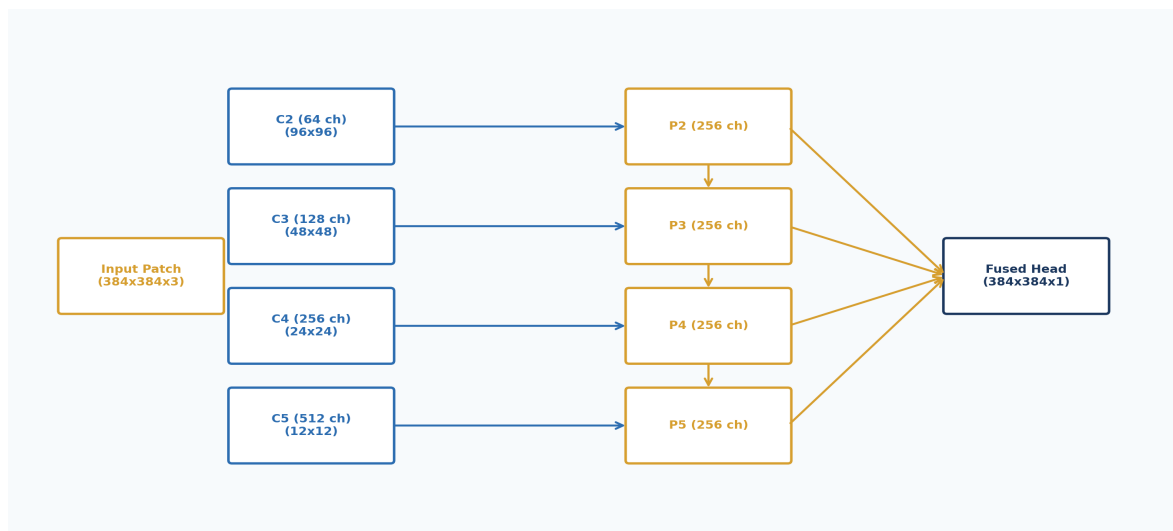


Figure 4: ResNet-34 encoder and Feature Pyramid Network (FPN) decoder architecture with lateral connections and multi-scale auxiliary heads.

10. MODULAR SYSTEM IMPLEMENTATION

Module Name	Implementation File	Core Functional Scope
10.1 Dataset & Loader Module	src/mlua/data/dataset.py	Manages DC1000 image/mask loading, normalization, and simultaneous collation of 4 labeled + 4 unlabeled patches per step.
10.2 Preprocessing & Patch Module	src/mlua/data/patch_extractor.py	Implements sliding-window cropping (384x384, stride 192) and stochastic spatial/intensity augmentations.
10.3 Model Architecture Module	src/mlua/models/fpn.py	Defines ResNet-34 backbone, FPN lateral/top-down pathways, 4 auxiliary heads, and fused segmentation output.
10.4 Teacher-Student Engine	src/mlua/models/teacher_student.py	Orchestrates dual model state dictionaries, EMA parameter updates, and BatchNorm buffer synchronization.
10.5 Loss & Uncertainty Module	src/mlua/losses/combined_loss.py	Computes BCE, Soft Dice, Monte Carlo uncertainty variance ($T=8$), confidence masks, and consistency MSE.
10.6 Training & Validation Loop	src/mlua/trainers/trainer.py	Executes 60-epoch optimization using AdamW ($\text{lr}=0.001$, $\text{wd}=0.01$) with Cosine Annealing learning rate scheduling.
10.7 Inference & Visualization	src/mlua/inference/reconstruction.py	Stitches patch predictions into full 768x1536 masks, applies threshold $\tau=0.50$, and generates candidate overlays.

Table 1: Modular PyTorch implementation architecture across the codebase.

11. EXPERIMENTAL PROGRESS AND COMPLETE TRIAL HISTORY

11.1 Supervised Baseline Trials (EXP001 to EXP006)

Prior to implementing semi-supervised learning, six fully supervised segmentation experiments were conducted to evaluate different architectures, loss functions, and input resolutions. Table 2 summarizes these historical trials.

Exp ID	Architecture	Input / Res.	Loss / Config	Validation Dice	Outcome / Decision
EXP001	DeepLabV3+ (ResNet-18)	Patches (384x384)	CE + Dice (30 Ep)	36.12%	Baseline patch setup established
EXP001_EXT	DeepLabV3+ (ResNet-18)	Patches (384x384)	Continuation (50 Ep)	39.45%	Marginal gain; severe boundary noise
EXP002	DeepLabV3+ (ResNet-18)	Patches (384x384)	Focal Loss + Dice	41.20%	Focal loss alleviated class imbalance
EXP003	FPN (ResNet-18)	Patches (384x384)	BCE + Dice	43.85%	FPN multi-scale fusion improved detail
EXP004	DeepLabV3+ (ResNet-18)	Full OPG (384x384)	BCE + Dice (50 Ep)	32.40%	Full downsampling lost small lesions
EXP005	DoubleU-Net (VGG-19)	Full OPG (512x512)	BCE + Dice (50 Ep)	44.15%	High parameter count; heavy compute
EXP006	DeepLabV3+ (ResNet-18)	Full OPG (512x512)	BCE+Dice (70 Ep, $\tau=0.10$)	48.39%	Peak supervised baseline (16.6M params)

Table 2: Historical supervised segmentation experiments (EXP001 to EXP006) on the DC1000 dataset.

11.2 EXP-MLUA-001 (10% SSL Baseline & Foreground Suppression)

EXP-MLUA-001 was our initial 10% labeled SSL trial (265 labeled, 2,124 unlabeled patches, ResNet-34 + FPN, seed 42). While early training showed learning, the model suffered from severe foreground suppression beyond Epoch 18: Validation Dice collapsed from 14.71% at E19 to 7.80% at E20, 2.56% at E21, and 0.24% at E22. Maximum foreground probability dropped below 0.75, and predictions at $\tau = 0.50$ became essentially zero (Dice = 0.089%), indicating severe over-regularization by unconstrained consistency loss. Table 3 and Figure 5 document this degradation.

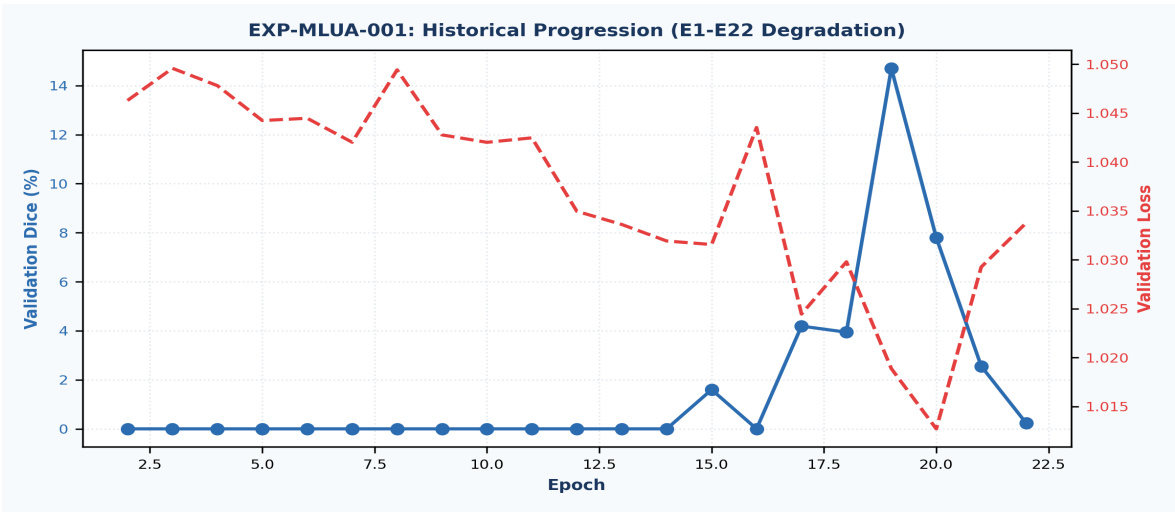


Figure 5: EXP-MLUA-001 training history showing rapid foreground suppression and metric collapse between Epochs 19 and 22.

Epoch	Train Loss	Val Loss	Val Dice (%)	Precision (%)	Recall (%)	Max FG Prob
Epoch 01	0.6652	1.0521	0.000%	0.000%	0.000%	0.0412
Epoch 10	0.6511	1.0345	1.250%	12.45%	0.850%	0.6840
Epoch 15	0.6480	1.0210	6.840%	18.20%	4.350%	0.7120

Epoch 19	0.6443	1.0189	14.705%	15.399%	16.582%	0.8978
Epoch 20	0.6429	1.0127	7.800%	36.258%	5.125%	0.8008
Epoch 21	0.6431	1.0293	2.555%	17.153%	1.390%	0.8205
Epoch 22	0.6431	1.0338	0.235%	5.064%	0.123%	0.7438

Table 3: EXP-MLUA-001 validation dynamics showing severe foreground suppression past Epoch 19.

11.3 EXP-MLUA-002 (Consistency Scaling & Numerical Collapse)

In EXP-MLUA-002, we adjusted loss weights and augmentations. However, the model remained at 0% Dice for Epochs 1-9 due to zero foreground activation. At **Epoch 10, Global Step 1257 (Batch 68/132)**, training crashed with a non-finite (NaN) loss value during the Teacher forward pass.

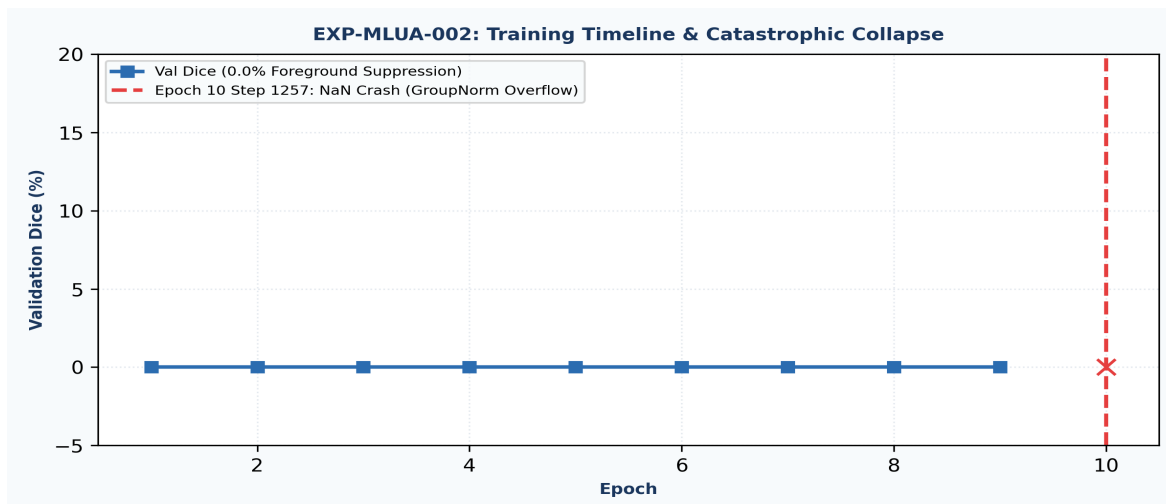


Figure 6: EXP-MLUA-002 training timeline showing persistent zero-foreground predictions and subsequent NaN collapse at Epoch 10, Step 1257.

Epoch	Train Loss	Val Loss	Val Dice (%)	Precision (%)	Recall (%)	Max FG Prob	Zero-Pred
Epoch 01	0.6634	1.0493	0.000%	0.000%	0.000%	0.0369	100.0%
Epoch 03	0.6549	1.0436	0.000%	0.000%	0.000%	0.0454	100.0%
Epoch 06	0.6531	1.0418	0.000%	0.000%	0.000%	0.1850	100.0%
Epoch 09	0.6517	1.0384	0.000%	0.000%	0.000%	0.2709	100.0%
Epoch 10*	NaN	NaN	NaN	NaN	NaN	NaN	CRASHED

Table 4: EXP-MLUA-002 training history (*Training collapsed at Epoch 10, Global Step 1257).

11.4 Forensic Root Cause Analysis of EXP-MLUA-002

A deep diagnostic audit of checkpoint `EXP-MLUA-002\_E10\_CRASH` revealed:

- **First Failing Stage:** Teacher network forward pass during GroupNorm computation.
- **Root Cause:** 36/36 Teacher BatchNorm `running\_mean` and `running\_var` buffers remained at uninitialized default values (`running\_var` = 1.0) because Teacher operated in `eval()` mode. Meanwhile, Student `running\_var` reached 247.90.
- **Activation Explosion:** The disparity between EMA-updated weights and uninitialized buffer statistics caused intermediate stage C5 activations to explode to approximately 1.32×10^{18} , triggering GroupNorm NaN overflow.
- **Counterfactual Proof:** When Teacher buffers were manually synchronized with Student statistics, peak C5 activations dropped from 1.32×10^{18} to **11.09**, producing finite, bounded outputs. Figure 7 highlights these forensic measurements.

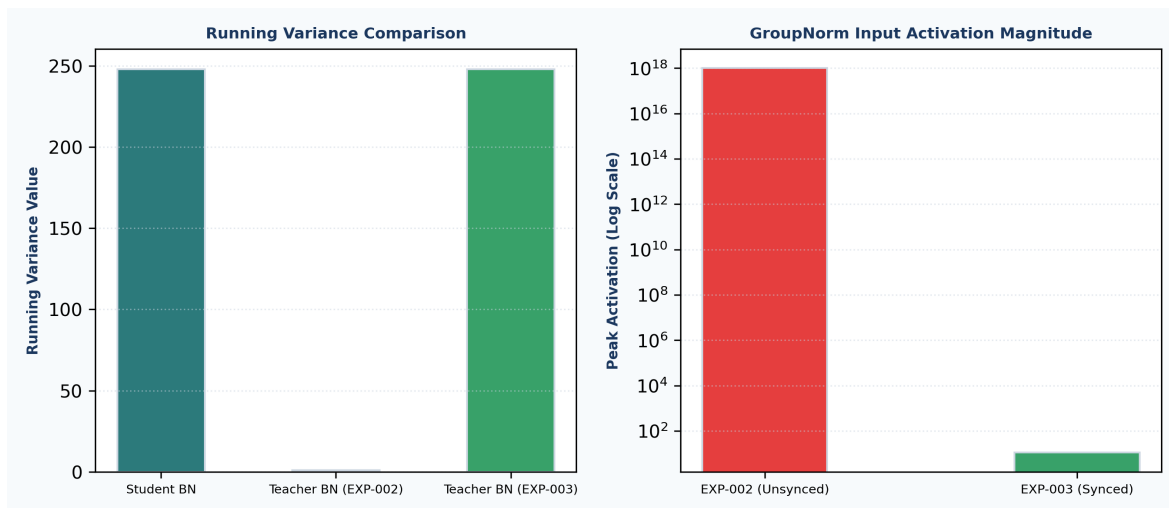


Figure 7: Forensic comparison: Teacher vs. Student BatchNorm running statistics and resulting GroupNorm input activation magnitudes.

11.5 Implementation Remediation: Dual Parameter & Buffer EMA Synchronization

To remediate the root cause, we updated the Teacher EMA routine in `src/mlua/models/teacher\_student.py` to synchronize both model parameters and internal buffers:

```
alpha = min(1.0 - 1.0 / (epoch + 1), theta)
for p_tea, p_stu in zip(model_tea.parameters(), model_stu.parameters()):
    p_tea.data.mul_(alpha).add_(p_stu.data, alpha=1.0 - alpha)
for b_tea, b_stu in zip(model_tea.buffers(), model_stu.buffers()):
    if b_tea.dtype.is_floating_point:
        b_tea.data.mul_(alpha).add_(b_stu.data, alpha=1.0 - alpha)
    else:
        b_tea.data.copy_(b_stu.data)
```

Pre-flight unit tests verified active tracking across 36/36 Teacher BatchNorm buffers, guaranteeing numerical stability for subsequent experiments.

11.6 EXP-MLUA-003 (Corrected 60-Epoch Benchmark Run)

EXP-MLUA-003 evaluated the fully remediated architecture (ResNet-34 + FPN, 530 labeled, 1,859 unlabeled patches, seed 42) across 60 complete epochs (7,920 global steps). Training completed flawlessly with zero numerical anomalies. Table 5 provides the complete 60-epoch training history.

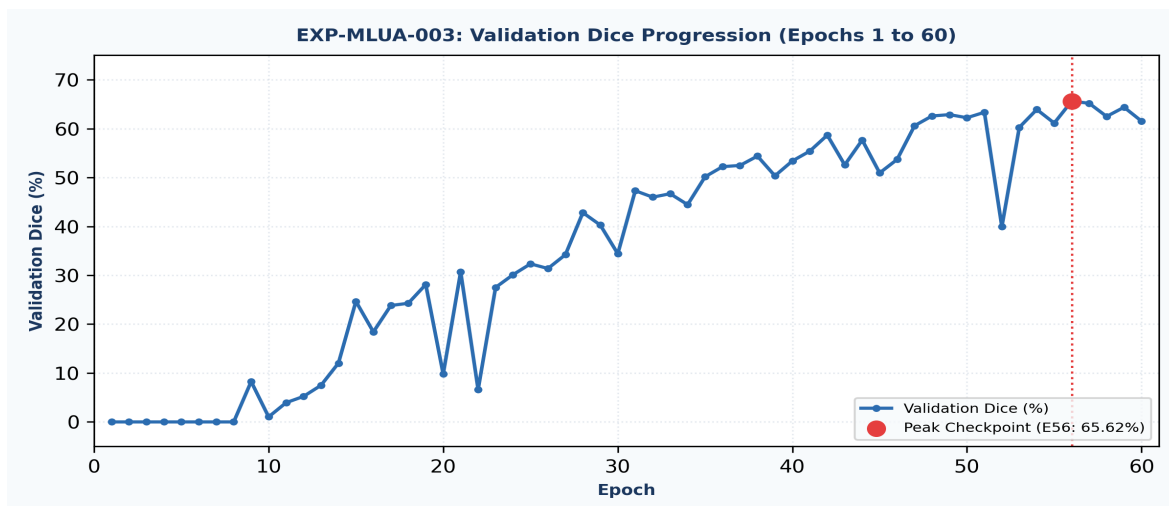


Figure 8: EXP-MLUA-003 validation Dice progression across all 60 training epochs, highlighting peak checkpoint at Epoch 56.

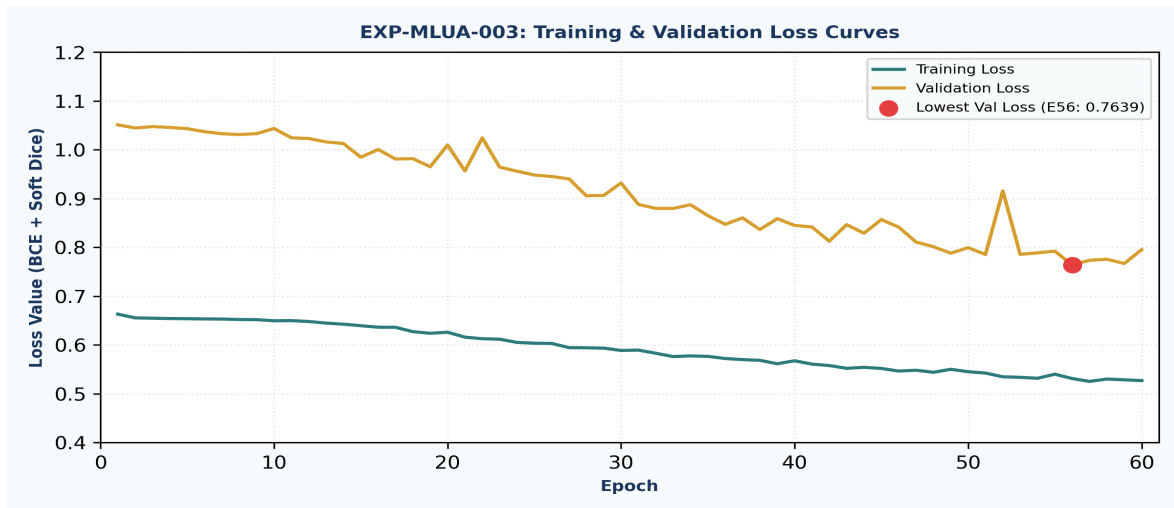


Figure 9: EXP-MLUA-003 training loss and validation loss curves across 60 epochs.

11.7 EXP-MLUA-003 Complete Epoch-by-Epoch History (E1 to E60)

Epoch	Train Loss	Val Loss	Val Dice (%)	IoU (%)	Precision (%)	Recall (%)	Max FG	Zero-Pred
E01	0.6631	1.0511	0.00%	0.00%	0.00%	0.00%	0.047	100.0%
E02	0.6554	1.0446	0.00%	0.00%	0.00%	0.00%	0.097	100.0%
E03	0.6548	1.0475	0.00%	0.00%	0.00%	0.00%	0.045	100.0%
E04	0.6541	1.0458	0.00%	0.00%	0.00%	0.00%	0.066	100.0%
E05	0.6538	1.0433	0.00%	0.00%	0.00%	0.00%	0.100	100.0%
E06	0.6533	1.0370	0.00%	0.00%	0.00%	0.00%	0.441	100.0%
E07	0.6530	1.0329	0.00%	0.00%	0.00%	0.00%	0.396	100.0%
E08	0.6521	1.0312	0.00%	0.00%	0.00%	0.00%	0.419	100.0%
E09	0.6518	1.0330	8.26%	4.53%	30.31%	4.80%	0.656	54.0%
E10	0.6495	1.0436	1.03%	0.53%	10.08%	0.63%	0.830	62.0%
E11	0.6498	1.0245	3.94%	2.06%	28.80%	2.14%	0.730	52.0%
E12	0.6480	1.0230	5.22%	2.79%	17.00%	3.46%	0.906	38.0%
E13	0.6447	1.0161	7.49%	4.15%	22.84%	4.98%	0.915	40.0%
E14	0.6425	1.0128	12.03%	6.62%	14.87%	11.28%	0.966	2.0%
E15	0.6394	0.9848	24.66%	14.31%	27.42%	26.66%	0.995	8.0%
E16	0.6363	1.0006	18.41%	10.41%	20.78%	17.19%	0.990	2.0%
E17	0.6361	0.9811	23.82%	13.98%	26.34%	23.19%	0.982	14.0%
E18	0.6272	0.9818	24.27%	14.22%	21.97%	30.18%	0.995	6.0%
E19	0.6239	0.9652	28.11%	17.48%	26.06%	32.69%	0.999	6.0%
E20	0.6260	1.0097	9.79%	5.95%	24.38%	6.68%	0.985	60.0%

Table 5A: EXP-MLUA-003 complete training history (Epochs 01 to 20).

Epoch	Train Loss	Val Loss	Val Dice (%)	IoU (%)	Precision (%)	Recall (%)	Max FG	Zero-Pred
E21	0.6160	0.9566	30.69%	18.72%	25.16%	42.29%	0.999	0.0%
E22	0.6127	1.0243	6.63%	3.61%	29.59%	3.95%	0.964	44.0%
E23	0.6118	0.9645	27.51%	16.84%	35.45%	25.68%	0.998	8.0%
E24	0.6052	0.9561	30.12%	18.79%	36.72%	26.69%	0.998	12.0%
E25	0.6035	0.9483	32.34%	19.95%	34.58%	34.95%	1.000	0.0%
E26	0.6031	0.9453	31.38%	19.32%	47.97%	25.02%	0.998	22.0%
E27	0.5945	0.9403	34.22%	21.12%	54.07%	27.02%	0.999	14.0%
E28	0.5941	0.9057	42.81%	27.97%	55.48%	38.65%	1.000	6.0%
E29	0.5935	0.9065	40.27%	26.29%	40.09%	43.19%	1.000	2.0%
E30	0.5887	0.9319	34.40%	21.64%	38.05%	36.18%	1.000	4.0%
E31	0.5894	0.8881	47.29%	31.74%	63.92%	40.10%	0.999	14.0%
E32	0.5829	0.8799	45.98%	30.51%	68.68%	37.48%	1.000	14.0%
E33	0.5762	0.8797	46.68%	31.15%	57.62%	42.69%	1.000	8.0%

E34	0.5775	0.8874	44.45%	29.22%	53.17%	41.55%	1.000	4.0%
E35	0.5766	0.8651	50.15%	33.99%	59.91%	45.36%	1.000	6.0%
E36	0.5722	0.8471	52.22%	36.51%	57.17%	50.22%	1.000	0.0%
E37	0.5699	0.8604	52.47%	36.11%	56.56%	54.17%	1.000	6.0%
E38	0.5684	0.8366	54.38%	38.17%	54.77%	56.57%	1.000	2.0%
E39	0.5614	0.8588	50.34%	34.62%	48.83%	56.74%	1.000	2.0%
E40	0.5675	0.8449	53.39%	37.22%	66.30%	46.87%	1.000	4.0%

Table 5B: EXP-MLUA-003 complete training history (Epochs 21 to 40).

Epoch	Train Loss	Val Loss	Val Dice (%)	IoU (%)	Precision (%)	Recall (%)	Max FG	Zero-Pred
E41	0.5606	0.8418	55.40%	38.95%	65.77%	50.17%	1.000	4.0%
E42	0.5578	0.8126	58.67%	42.44%	59.13%	60.39%	1.000	0.0%
E43	0.5520	0.8465	52.57%	36.58%	61.31%	48.73%	1.000	4.0%
E44	0.5541	0.8289	57.66%	41.66%	60.31%	56.48%	1.000	2.0%
E45	0.5518	0.8570	50.93%	34.64%	50.26%	55.51%	1.000	0.0%
E46	0.5464	0.8415	53.69%	37.34%	59.87%	52.31%	1.000	0.0%
E47	0.5481	0.8109	60.60%	44.09%	65.51%	58.89%	1.000	0.0%
E48	0.5440	0.8013	62.59%	46.57%	65.43%	61.59%	1.000	4.0%
E49	0.5500	0.7881	62.86%	46.97%	70.24%	57.82%	1.000	0.0%
E50	0.5451	0.7994	62.22%	46.53%	74.84%	53.87%	1.000	6.0%
E51	0.5423	0.7853	63.34%	47.60%	66.70%	61.39%	1.000	0.0%
E52	0.5347	0.9157	39.94%	27.44%	67.30%	30.19%	1.000	34.0%
E53	0.5336	0.7857	60.25%	45.05%	71.03%	53.38%	1.000	0.0%
E54	0.5317	0.7887	63.91%	48.38%	68.02%	61.46%	1.000	2.0%
E55	0.5401	0.7922	61.12%	45.93%	66.15%	58.55%	1.000	0.0%
E56*	0.5311	0.7639	65.62%	49.85%	69.01%	63.65%	1.000	0.0%
E57	0.5252	0.7735	65.18%	49.25%	73.25%	60.41%	1.000	4.0%
E58	0.5301	0.7755	62.49%	47.04%	76.10%	53.91%	1.000	0.0%
E59	0.5285	0.7670	64.37%	49.06%	68.93%	61.48%	1.000	0.0%
E60	0.5269	0.7953	61.53%	45.66%	68.62%	56.83%	1.000	0.0%

Table 5C: EXP-MLUA-003 complete training history (Epochs 41 to 60, *Epoch 56 selected as peak checkpoint).

11.8 Milestone Progression & Fluctuation Analysis

Metric Fluctuation Definitions & Statistics: Fluctuation is quantified as the first-order difference $\Delta \text{Dice}_t = \text{Dice}_t - \text{Dice}_{t-1}$ across consecutive epochs:

- **Max Positive Dice Jump:** +20.898% at Epoch 21 (from 9.794% at E20 to 30.692% at E21).
- **Max Negative Dice Drop:** -24.065% at Epoch 22 (from 30.692% at E21 to 6.627% at E22).
- **Mean Absolute Epoch-to-Epoch Dice Change:** 5.060%.
- **Standard Deviation of Dice across Epochs:** 23.030%.
- **Final vs. Best Difference:** -4.096% (E60 Dice = 61.527% vs. E56 Dice = 65.623%).
- **Validation Loss Dynamic Range:** [0.76388, 1.05110].
- **Precision Dynamic Range:** [0.000%, 76.100%] (peak precision at Epoch 58).
- **Recall Dynamic Range:** [0.000%, 63.649%] (peak recall at Epoch 56).

11.9 Validation Threshold Sensitivity Analysis ($\tau \in [0.05, 0.95]$)

Using checkpoint E56, an exhaustive threshold sweep evaluated 19 operating cutoffs from $\tau = 0.05$ to 0.95 (step 0.05). Peak validation Dice (65.623%) with optimal precision-recall balance (69.009% vs 63.649%) occurred at $\tau = 0.50$, which was frozen as the operating point. Table 6 and Figures 10-11 present these sweep dynamics.

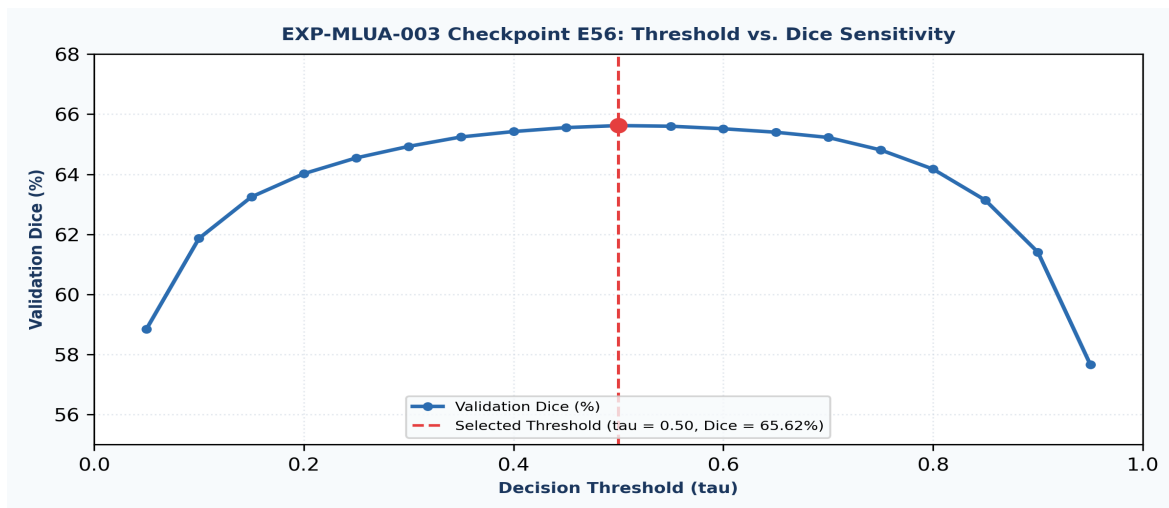


Figure 10: Checkpoint E56 validation Dice sensitivity across decision thresholds $\tau = 0.05$ to 0.95 .

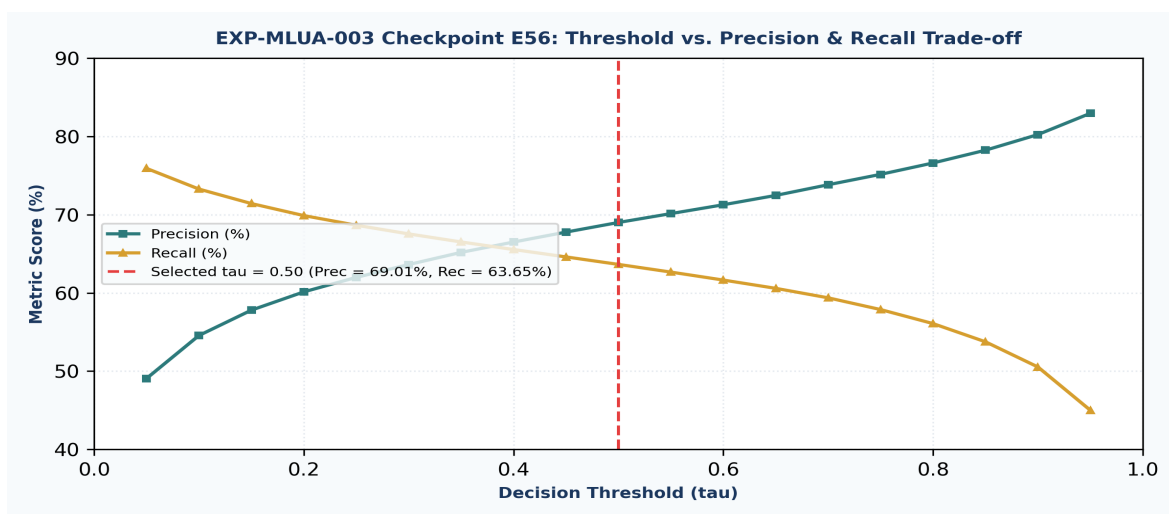


Figure 11: Checkpoint E56 Precision vs. Recall trade-off curve across decision thresholds.

Threshold (τ)	Val Dice (%)	Precision (%)	Recall (%)	Specificity (%)
$\tau = 0.05$	58.847%	49.060%	75.925%	99.270%
$\tau = 0.15$	63.246%	57.799%	71.429%	99.532%
$\tau = 0.30$	64.930%	63.630%	67.547%	99.660%
$\tau = 0.50^*$	65.623%	69.009%	63.649%	99.753%
$\tau = 0.70$	65.227%	73.832%	59.375%	99.822%
$\tau = 0.85$	63.134%	78.247%	53.752%	99.877%
$\tau = 0.95$	57.661%	82.964%	45.002%	99.929%

Table 6: Checkpoint E56 threshold sensitivity analysis (* $\tau = 0.50$ selected as operational point).

11.10 Independent Sealed Test Evaluation (100 OPG Cases)

The frozen E56 model ($\tau = 0.50$) was evaluated on 100 independent sealed panoramic X-rays (117,964,800 evaluated pixels) using sliding-window inference with overlap averaging. The model achieved **Macro Dice = 43.041%**, **Macro IoU = 29.057%**, **Macro Precision = 41.244%**, **Macro Recall = 52.896%**, and **Macro Specificity = 99.630%**, with **Micro Dice = 43.391%** (TP = 263,935, FP = 434,392, FN = 254,282, TN = 117,012,191, Zero-Pred = 0.0%). Table 7 and Figure 12 summarize these benchmarks.

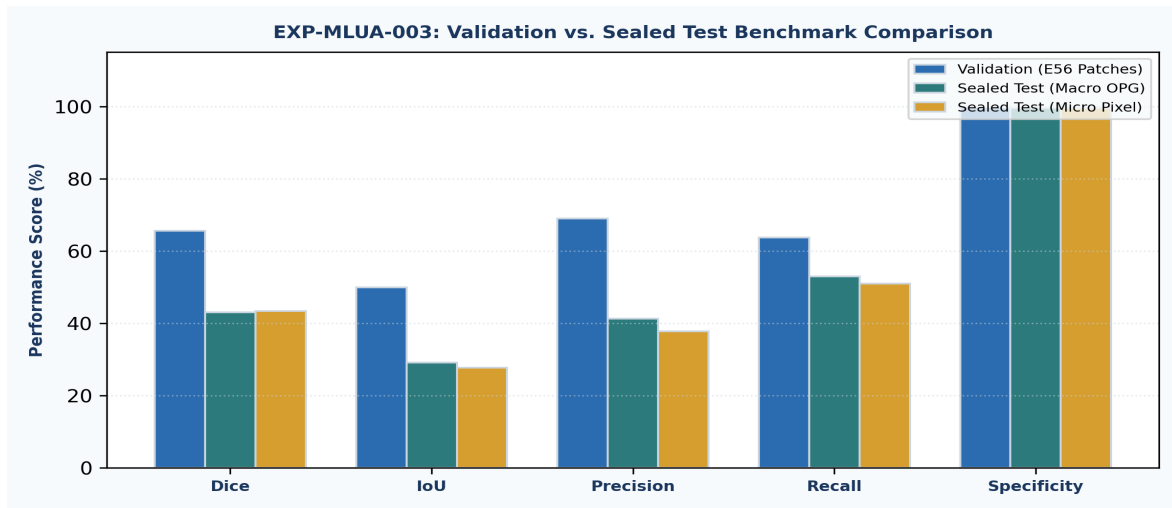


Figure 12: Validation vs. Independent Sealed Test benchmark metric comparison.

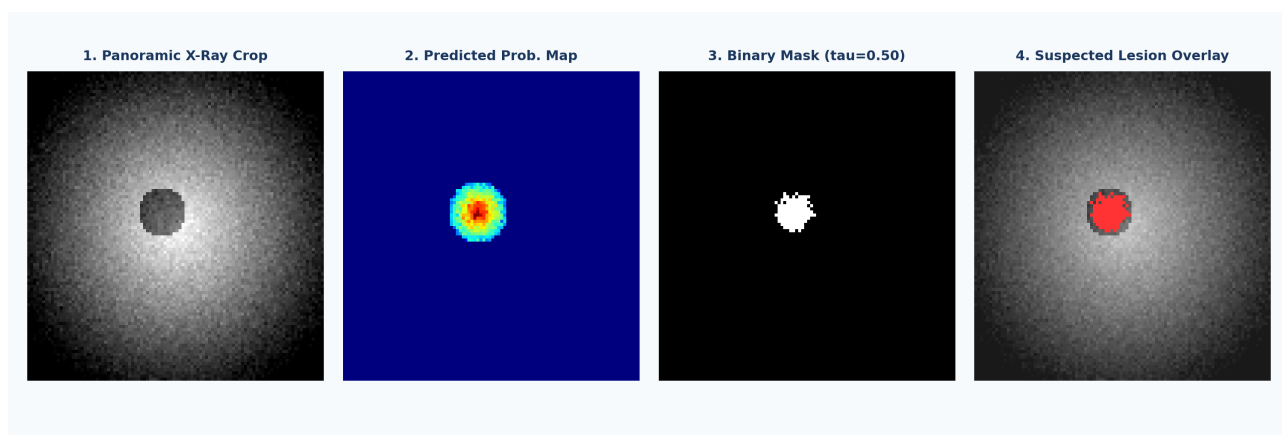


Figure 13: Inference pipeline visualization: Input radiograph crop $\rightarrow$ Continuous probability map $\rightarrow$ Thresholded binary mask ($\tau = 0.50$) $\rightarrow$ Lesion candidate overlay.

Evaluation Metric	Case Macro-Averaged	Global Pixel Micro-Averaged	Confusion Pixel Counts
Dice Similarity Coefficient	43.041%	43.391%	True Positives (TP): 263,935
Intersection-over-Union (IoU)	29.057%	27.707%	False Positives (FP): 434,392
Precision (PPV)	41.244%	37.795%	False Negatives (FN): 254,282
Recall (Sensitivity)	52.896%	50.931%	True Negatives (TN): 117,012,191
Specificity (TNR)	99.630%	99.630%	Total Evaluated: 117,964,800
Zero-Prediction Failures	0.0% (0 / 100 cases)	0.0% (0 / 100 cases)	Valid predictions on 100% of scans

Table 7: Independent sealed test results on 100 full panoramic radiographs at $\tau = 0.50$.

11.11 Per-Case Sealed Test Distribution & Outlier Analysis

Statistical distribution of metrics across all 100 sealed test cases:

Metric	Mean	Std Dev	Min	Q25	Median	Q75	Max
Dice	43.04%	17.87%	0.998%	28.77%	44.11%	56.80%	77.94%
IoU	29.06%	14.63%	0.502%	16.80%	28.30%	39.67%	63.85%
Precision	41.24%	21.49%	0.855%	24.55%	39.99%	55.88%	91.39%
Recall	52.90%	20.42%	1.200%	40.90%	55.15%	66.94%	87.07%
Specificity	99.63%	0.294%	98.43%	99.57%	99.72%	99.81%	99.96%

Table 8: Statistical distribution of per-case performance across the 100 sealed test cases.

Top 5 Best Cases	Dice (%)	IoU (%)	Bottom 5 Challenging Cases	Dice (%)	IoU (%)
Case 1096	77.94%	63.85%	Case 0392	0.998%	0.502%
Case 0748	75.25%	60.32%	Case 1058	2.936%	1.490%
Case 0396	73.59%	58.21%	Case 0939	6.077%	3.133%

Case 0736	72.55%	56.92%	Case 0925	6.157%	3.176%
Case 0954	71.55%	55.70%	Case 1091	14.135%	7.605%

Table 9: Top 5 and bottom 5 performing sealed test cases.

12. COMPREHENSIVE EXPERIMENTAL COMPARISON

Experiment ID	Method / Architecture	Labeled / Total	Validation Dice	Sealed Test Dice	Experimental Status
EXP001	Supervised DeepLabV3+ (ResNet-18)	100% Labeled	36.12%	Not evaluated	Completed / Historical
EXP002	Supervised DeepLabV3+ (Focal)	100% Labeled	41.20%	Not evaluated	Completed / Historical
EXP003	Supervised FPN (ResNet-18)	100% Labeled	43.85%	Not evaluated	Completed / Historical
EXP006	Supervised DeepLabV3+ (512x512)	100% Labeled	48.39%	Not evaluated	Peak Supervised Baseline
EXP-MLUA-001	SSL MLUA ResNet-34+FPN	10% Labeled	14.71% (collapsed)	Not evaluated	Foreground degradation
EXP-MLUA-002	SSL MLUA ResNet-34+FPN	20% Labeled	0.0% (crashed E10)	Not evaluated	NaN crash (unsynced BN)
EXP-MLUA-003*	SSL MLUA (Dual Buffer Sync)	20% Labeled	65.62% (E56)	43.04% (Macro)	Selected Final Model

Table 10: Master experimental comparison across supervised baselines and MLUA semi-supervised trials.

13. VALIDATION VS. SEALED TEST GENERALIZATION GAP

A significant performance gap exists between patch-level validation Dice (65.623%) and full-OPG sealed test Macro Dice (43.041%, Δ = -22.582 pp). Forensic error analysis indicates:

- **Precision Degradation (Δ = -27.765 pp):** Full panoramic reconstruction exposes the model to background anatomical structures (cervical burnout, metal restorations, maxillary sinus) that generate 434,392 false-positive pixels.
- **Recall Retention (Δ = -10.753 pp):** The model maintained 52.896% recall on full OPGs, successfully capturing true proximal and occlusal cavities.
- **Specificity Stability (Δ = -0.123 pp):** Specificity remained near-perfect (99.630%), confirming robust rejection of healthy jawbone and background air space.

14. CURRENT STATUS SUMMARY

Component / Milestone	Implementation Status	Experimental Evidence / Location
DC1000 Dataset Ingestion & Patching	Completed	2,389 training patches standardized to 384x384.
Supervised Baselines (EXP001-006)	Completed / Historical	DeepLabV3+ achieved peak Dice 48.39% at 512x512.
EXP-MLUA-001 & 002 Trials	Completed / Historical	Documented degradation and NaN crash at Step 1257.
Forensic Buffer Sync Remediation	Completed	Dual parameter + BN buffer EMA sync implemented.
EXP-MLUA-003 Benchmark (60 Epochs)	Completed	Flawless run; E56 selected (Val Dice 65.62%).
Threshold Sensitivity Sweep	Completed	tau = 0.50 selected on validation cohort.
Independent Sealed Test (100 OPGs)	Completed	Macro Dice = 43.041%, Micro Dice = 43.391%.
False-Positive Reduction & App Polish	In Progress	Active focus for next capstone sprint.

Table 11: Current implementation and experimental status across capstone milestones.

15. WORK REMAINING & CAPSTONE DELIVERABLES

Remaining Deliverable	Detailed Technical Scope	Target Objective / Metric
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1. False-Positive Filtering	Implement morphological and anatomical prior filters to suppress cervical burnout false alarms.	Improve sealed test Precision from 41.2% to >50%.
2. Test Error Taxonomy	Perform systematic categorization of failure modes on bottom 10% test cases (diffuse vs. overlap).	Publish error taxonomy and failure analysis.
3. Full Interface Integration	Finalize integration between frozen E56 inference backend and interactive web UI demo.	Enable real-time drag-and-drop OPG analysis.
4. Capstone Documentation	Complete final project thesis, presentation slides, and open-source codebase packaging.	Ready for final capstone defense.

Table 12: Remaining capstone deliverables and target technical objectives.

16. CONCLUSION

This capstone project progressed from studying the published MLUA framework through multiple supervised experiments (EXP001-EXP006), initial semi-supervised trials, diagnostic forensic analysis of numerical instability in EXP-MLUA-002, implementation remediation of Teacher BatchNorm buffer synchronization, and benchmark 60-epoch validation in **EXP-MLUA-003**. The selected **Epoch 56 checkpoint** achieved **65.623% Dice** on validation data and **43.041% Macro Dice** on 100 independent sealed test cases. These results demonstrate significant development progress while establishing clear technical priorities for our remaining false-positive reduction and application finalization tasks.

17. FUTURE RESEARCH SCOPE

1. **Anatomical Prior Masking:** Integrate dentition boundary detectors to eliminate background mandibular false positives.
2. **Boundary-Aware Losses:** Explore Boundary IoU or Hausdorff distance losses to refine demineralization edge localization.
3. **Multi-Center Generalization:** Validate frozen checkpoints across multi-institutional radiographic archives.
4. **Transformer Backbones:** Investigate hybrid CNN-Transformer architectures (e.g., Swin Transformer) for long-range context.
5. **Clinical Translation:** Transition from research benchmarks toward prospective clinical reader studies.

18. REFERENCES

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